# PHIL1002: Introduction to Critical Thinking

Lecture 3B: Argument Evaluation-I

Support and Validity

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## Agenda: Argument Evaluation

Part 1 — Support and Validity

Part 2 — Truth and acceptability

Part 3 — Cogency and Soundness

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# Support

***Truth*** is a property of the statements given as premises: roughly, a statement is true if the world actually is as the statement claims it to be.

Support is **not** a property **of** statements, but a **relation between statements**, a relation which we claim exists when we use some statements as premises and argue to others as conclusions.

A strong argument should have premises which support its conclusion.

The crucial things to keep in mind are **that** truth and support are different, and **how** they are different.

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# Support and truth

To see what the difference is, consider this analogy.

Suppose you have to rest a heavy object safely on a beam.

You need to consider the weight of the object, the strength of the beam, and the security of the beam's fixings.

A strong beam is no use without secure fixings, and secure fixings are no use with a weak beam.

But these two factors are **independent of each other** so that **one can hold while the other does not.**

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# Two separate questions

There are two questions to consider when evaluating an argument:

**(1) Are the premises true?**

And,

**(2) Whether or not they're true, how much support do they give the conclusion?**

Today we focus on the second question.

This is not a question about the premises but about the **relation** between premises and conclusion.

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# Support

The support a statement gives to a conclusion can be strong even if the statement is known to be false.

To find out how much support the premise gives to the conclusion, you have to assume that the premise is true.

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# Assessing support

**'Supposing for the sake of argument that these premises are true, how probable does this make it that the conclusion is true too?'**

**'Supposing for the sake of argument that these premises are true, how *<u>im</u>probable* does this make it that the conclusion is *false*?'**

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# Complete support

Govier calls the notion of complete support 'deductive entailment'. (AKA: 'deductive validity', often simply shortened to 'validity'.)

**From now on you must use the word 'valid' only with this meaning.**

In particular, **never** say that statements or premises are valid. Statements are true or false.

It's only **arguments** which can be valid or invalid.

Conversely, arguments cannot be true or false, because they are not statements but **relations between statements**.

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# Complete support

Complete support between premises and conclusions, that is, deductive validity of an argument is defined this way:

***An argument's being deductively valid = The truth of the premises' guaranteeing absolutely the truth of the conclusion***

What if the premises are false? The definition would still apply.

The premises may not in fact **be** true, but **if they were true**, they would still guarantee absolutely the truth of the conclusion.

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# A valid argument…consisting entirely of falsehoods:

*Napoleon Bonaparte died in 1902;*

*therefore*

*Napoleon Bonaparte was alive in the twentieth century.*

The truth of the premise would guarantee absolutely the truth of the conclusion even though both of them are **in fact** false.

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# Deductive validity

A certain kind of impossibility is required for deductive validity.

This stronger sort is ***logical impossibility.***

Something is logically impossible if the attempt to describe it involves a ***contradiction***, which is what you get when you both assert and deny the same thing, at the same time.

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# Contradiction

A contradiction may be **explicit**, like this one:

*The earth goes round the sun and the earth does not go round the sun*

where the same statement is visibly both asserted and denied.

Or it may be **implicit** (as in the previous argument):

*Napoleon died in 1902 and Napoleon was not alive in the twentieth century*

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# Contradiction

***Napoleon died in 1902 and Napoleon was not alive in the twentieth century***

How do I get this contradiction? I got it by putting the premises of the argument and the ***negation*** of the conclusion together.

***Napoleon Bonaparte died in 1902;***

*therefore*

***Napoleon Bonaparte was alive in the twentieth century.***

If this gives a contradiction, then the argument you're examining is deductively valid.

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# The test for validity

1. Drop inference indicators.
2. Conjoin (&) premises and negation of conclusion (i.e. suppose that the premises are true, and conclusion is false).
3. If result is a **contradiction**, argument is **valid**.

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# What about an implied contradiction?

The way to test for this is to see if it is reducible to an explicit contradiction by definition:

'die' = 'pass from being alive to being not alive'

'twentieth century' = 'the years 1901 and 1902 and 1903 … and 2000'

***Napoleon was alive in 1902 and Napoleon was not alive in 1902***

**The appearance of this contradiction is the final proof that that argument is valid.**

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# Degrees of support…

…which are less than complete.

The more unlikely it is that the conclusion could be false if the premises are true, the stronger the support.

Except for the two extremes of complete and zero support, it is only in rare or artificial cases that the probability can be quantified.

In most cases, we can do no more than classify the degree as complete, strong, moderate, weak, and nil.

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# Strong support

A premise whose truth would make the truth of the conclusion highly likely is a premise which gives that conclusion **strong support**.

This is not a matter of the premise's actually being true, but a matter of what the consequences of its truth would be if it **was** true.

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# An example...

*Napoleon Bonaparte was still young, fit and strong late on December 31st, 1900*

*so*

*Napoleon Bonaparte lived on into the twentieth century.*

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# Degrees of support

Consider this example:

*I have tossed this coin three times in succession, and it has come down heads each time;*

*so*

*the coin is biased.*

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# Estimating degrees of support

Ask yourself this key question:

***Supposing the premise(s) were true, is there any way not involving contradiction in which the conclusion nevertheless could be false?***

If there is no way – because it would involve a contradiction – in which the conclusion could be false if the premise(s) were true, then the degree of support is complete.

If you can think of some way(s) in which the conclusion could be false even if the premise(s) were true, then the support is **not** complete.

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# An example…

Suppose you're driving your car and it suddenly stops. You reason this way:

*The motor has stopped.*

*The temperature gauge indicates that it has greatly over-heated.*

*So*

*The motor must have seized.*

To know whether this diagnosis of the problem is correct, you need to be able to determine how strong the support is…

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# First, remember the method:

**DROP INFERENCE INDICATORS & NEGATE CONCLUSION.**

**IS THERE A CONTRADICTION?**

*The motor has stopped. The temperature gauge indicates that it has greatly over-heated. The motor has not seized.*

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# Second, ask the question…

'is it possible that the premises could true, and the conclusion false'?

Suppose we rule out all other causes of a motor's stopping so that we know that the motor really has greatly over-heated **and** that this has somehow caused it to stop.

This reasoning might be:

*The motor has stopped. It has greatly over-heated. The over-heating caused it to stop. So it must have seized.*

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# Estimating degrees of support

A rule of thumb:

***the more realistic the possibility of a situation in which the premises are true yet the conclusion false, then the weaker the support.***

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# Deductive validity

Although arguments can involve different degrees of support, the case of **complete support** is so important that it gets its own name:

**DEDUCTIVE VALIDITY**

Deductive validity is the basis of mathematical proof and symbolic or formal logic.

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# *Form*al logic

*All cows are mammals*

*All mammals are things which have hearts*

*Therefore*

*All cows are things which have hearts*

That looks like a good argument, with true premises which give complete support to their conclusion.

What accounts for the support which the premises give to the conclusion is what we can call the *form* of the argument itself.

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## *Form*al logic

*All C are M*

*All M are H*

*Therefore*

*All C are H*

Any argument of this form is deductively valid.

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## The first reason is…

….many syllogisms which one can't tell just by inspection whether they are valid, some mechanical method is necessary.

*All non-angels are non-dancers on the heads of pins*

*All dancers on the heads of pins are entrants into the Kingdom of Heaven*

*Therefore*

*No angels are non-entrants into the Kingdom of Heaven*

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# The second reason…

…Aristotle's example inspired the invention of far more powerful logical systems.

For example:

*Either the cat ate the canary, or Grannie did. But the cat didn't. So, Grannie did.*

This argument is valid: if the premises are true, then they guarantee the truth of the conclusion.

Any argument of the same form would be valid too.

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# Example…

Here's an argument with the same form:

*Either the logging is sustainable, or Department of Biodiversity,Conservation and Attractions is not telling us the truth.*

*Clearly the logging is unsustainable,*

*so*

*DBCA is not telling us the truth.*

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# Disjunctive syllogism

***Either P or Q***

***Not P***

***so,***

***Q***

(where the capital letters stand for any statements whatsoever.)

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# Formal Logic

**Syllogistic logic:**

- the letters stand for classes of objects.
- is a calculus of **terms**—we get inside the statements into the microstructure of the arguments.

**Propositional calculus:**

- the letters stand for whole statements.
- is restricted to the relations between whole statements.
- is unable to recognise any connections between statements which arise from their containing identical terms—deals with an argument's macrostructure.

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# Predicate Logic

A logical calculus which combines the strengths of both syllogistic and propositional calculus.

It's known as 'predicate calculus'.

It forms the basis of most of the programming languages used for artificial intelligence systems.

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# Form and content

Some inferences are legitimised by their **form** alone, others depend upon the **content** of what is said.

To see the difference, an example:

*Napoleon Bonaparte died in 1902*

*therefore*

*Napoleon Bonaparte was alive in the twentieth century.*

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# Form and content

Premises can **support** conclusions in at least two different ways.

***Purely formally***, or by means of the ***content*** of those premises.

Using formal calculi, you can't estimate the degree of support which premises give to conclusions by virtue of their content.

The method I've shown you today **does** enable you to do this.

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# Support

My approach to the topic of support is different from that which you will find if you go to standard introductory logic texts.

Traditional writers on logic treat validity as an all or nothing matter—an argument was either valid (meaning deductively valid) or invalid.

I make distinctions within the range of invalid arguments.

If an argument is invalid, then its degree of support for its conclusion is less than complete, but it may still be very strong.

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# Relevance and grounds

There is no principled reason for dividing relevance from grounds.

We've seen that support is a matter of degrees ranging from zero to complete.

A premise which is irrelevant to its conclusion merely gives that conclusion support of degree zero.

'Grounds' is the heading under which Govier discusses all other degrees of support.

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# Next lecture...

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